

A novel method for high-throughput phenotyping for dry root rot disease

Materials

1. High-throughput imaging system: RGB and thermal sensors.
2. DSLR Camera: Canon EOS 90D equipped with a 32.5-megapixel APS-C CMOS sensor and DIGIC 8 image processor, mounted on a custom-engineered, curved-roof-rail assembly.
3. Thermal Camera: Fluke TiX580 (Fluke Corporation, Everett, WA, USA) featuring 640 × 480 pixels infrared resolution and thermal sensitivity of $\leq 0.05^{\circ}\text{C}$ at 30°C, mounted on a manually operated gantry rail system with a height-adjusting mechanism.
4. Flatbed Scanner: Epson Perfection V850 Pro (Seiko Epson Corp., Nagano, Japan) paired with a transparent acrylic tray.
5. Reagents and Consumables: Distilled water for root suspension.
6. Analysis Software: WinRHIZO Pro 2022 (Regent Instruments Inc., Canada) and Fluke SmartView 4.4.363.0 software (Fluke thermography, Everett, WA, USA).

Methods

Aerial RGB imaging

1. Mount the Canon EOS 90D DSLR camera on the curved-roof-rail assembly at a fixed height to maintain a consistent ground sampling distance.
2. Standardize camera settings using a fixed ISO range (Standard ISO 100-25600) to ensure uniform color consistency across all images.
3. Move the camera assembly smoothly and manually along the length of the experimental plots to capture high-resolution images of the canopy.
4. Use the acquired images to monitor plant growth, canopy coverage, and morphological traits.

Thermal imaging

1. Mount the Fluke TiX580 thermal camera onto the gantry rail system located directly above the canopy level.
2. Adjust the height mechanism to position the sensor for close-range imaging while avoiding any physical disturbance to the plants.
3. Set the camera emissivity to 0.98 to match the radiant properties of green plant tissue.
4. Execute thermal imaging during the 50% flowering window from 12:00 pm to 02:00 pm, capturing data at the transition from the vegetative to reproductive stage.
5. Record canopy temperature and physiological stress signatures before visible DRR symptoms develop.

Root scanning

1. Uproot the plant root systems carefully to maintain structural integrity.
2. Wash the roots thoroughly with water to eliminate all soil debris.
3. Fill a transparent acrylic tray with a thin layer of distilled water and place it onto the Epson Perfection V850 Pro flatbed scanner.
4. Spread the washed roots gently within the water layer to minimize overlapping strands and scanning artifacts.
5. Scan the root systems at a resolution of 600 DPI in 24-bit color depth and save the high-resolution files in TIFF format.

Chlorosis measurement

1. Open the high-resolution aerial RGB images in the WinRHIZO Pro 2022 software.
2. Develop a specific Color Analysis Class (CAC) file by defining the Hue, Saturation, and Value (HSV) color spectra for target tissues.
3. Segment the plant canopy into two thematic classes: healthy (green) and chlorotic (yellow/pale-green).

4. Run the software to calculate the precise pixel surface area belonging to each defined color class.
5. Calculate the final percentage of disease-induced chlorosis relative to the total canopy area using the pixel values.

Disease severity index calculation

1. Import the 600 DPI root TIFF images into WinRHIZO Pro 2022.
2. Create a root-specific CAC file that segments the root tissue into five distinct color levels representing necrosis progression.
3. Assign the segmentation values from healthy tissue (Color class 0) through increasingly dark sections up to completely necrotic black tissue (Color class 4).
4. Extract the software-calculated surface area and total volume (N) associated with each user-defined color class.
5. Calculate the percent disease severity index by using the following formula

$$DSI (\%) = \frac{4A + 3B + 2C + 1D + 0E}{4N} \times 100$$

A, B, C, D, and E correspond to the area of each color classes

Canopy temperature depression

1. Import the collected infrared thermal images into the Fluke SmartView software.
2. Select and export the ambient and canopy temperatures from multiple representative regions or points within the images.
3. Calculate the difference between the ambient temperature and the canopy temperature to determine the canopy temperature depression (CTD).
4. Interpret the results noting that a higher CTD value indicates a healthier plant status.

Note: The images acquired from laboratory conditions and greenhouse conditions also can be processed through the same way.

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